
The Geometry of Pokémon: Linear Representations of Fictional Franchise Membership in GPT-2-XL

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Abstract

The Linear Representation Hypothesis posits that high-level concepts are encoded as directions in the representation spaces of large language models. While this has been demonstrated for properties like truth, sentiment, and language identity, no prior work has investigated whether membership in a specific fictional franchise is similarly represented. We extract a “Pokémon direction” from GPT-2-XL’s residual stream using the mass-mean probing technique and find that the concept “X is a Pokémon” is linearly encoded, achieving 88.4% classification accuracy ($AUC = 0.934$) on held-out entities—far above the 50.5% random baseline. By projecting 177 non-Pokémon entities onto this direction, we discover a continuous spectrum of “Pokémon quality”: invented Pokémon-sounding names, Nintendo game creatures (Cactuar, Koopa, Kirby), and Digimon score highest, while common real-world objects score lowest. The category-level differences are highly significant (Kruskal-Wallis $p = 1.03 \times 10^{-10}$). Thirteen non-Pokémon entities cross the classification boundary, including six game creatures and one Digimon. These results extend the Linear Representation Hypothesis to fine-grained fictional franchise membership, revealing that language models encode not a binary “is/isn’t Pokémon” distinction but a graded, interpretable measure of Pokémon-likeness.

1 Introduction

What makes something a Pokémon? To a human, the answer involves a mix of visual design, naming conventions, franchise knowledge, and cultural association. But how does a language model represent this concept? Recent work on the Linear Representation Hypothesis [Park et al., 2024] has shown that high-level concepts—truth and falsehood [Marks and Tegmark, 2024], language identity, and factual associations [Meng et al., 2022]—are encoded as linear directions in transformer representation spaces. We ask whether this extends to a category that is at once highly specific and culturally rich: membership in the Pokémon franchise.

This question matters for two reasons. First, fictional franchise membership is a novel test case for the Linear Representation Hypothesis. Unlike truth or sentiment, “is a Pokémon” is a culturally constructed category with fuzzy boundaries—Digimon look like Pokémon, many real animals inspired Pokémon designs, and invented creature names can sound convincingly Pokémon-like. If a simple linear direction can capture this category, it suggests that linear representations extend well beyond the clean binary concepts studied so far. Second, the question “which non-Pokémon entities are most Pokémon-like according to a language model?” is intrinsically interesting. It reveals how models organize knowledge about fictional entities and whether franchise boundaries are sharp or graded in representation space.

We use the mass-mean probing technique from Marks and Tegmark [2024] to extract a Pokémon direction from GPT-2-XL’s residual stream. We find that this direction achieves 88.4% classification accuracy on held-out entities ($AUC = 0.934$), substantially outperforming both a random direction baseline (50.5%) and a logistic regression probe (83.2%). The direction emerges as early as layer 4 and peaks in the middle layers (10–32), consistent with prior findings that conceptual representations are strongest in middle transformer layers [Meng et al., 2022, Marks and Tegmark, 2024].

By projecting 177 non-Pokémon entities across seven categories onto this direction, we uncover a continuous spectrum of Pokémon quality. Invented Pokémon-sounding names and Nintendo game creatures score highest; Digimon rank third; common real-world objects are maximally distant. The category-level ordering is highly significant (Kruskal-Wallis $H = 58.22$, $p = 1.03 \times 10^{-10}$) and matches human intuitions about Pokémon-likeness. Thirteen non-Pokémon entities cross the classification boundary, including Cactuar, Koopa, Kirby, and the Digimon Garurumon.

In summary, our main contributions are:

- We demonstrate that fictional franchise membership (“is a Pokémon”) is linearly encoded in GPT-2-XL’s residual stream, extending the Linear Representation Hypothesis to a new and culturally rich category.
- We discover that non-Pokémon entities exhibit a continuous, interpretable spectrum of Pokémon quality, with game creatures and invented monster names scoring highest and common objects scoring lowest.
- We show that the mass-mean probe outperforms logistic regression for this task (88.4% vs. 83.2%), consistent with prior findings on the superior causal meaningfulness of unsupervised directions.

2 Related Work

Linear representations in language models. The Linear Representation Hypothesis (LRH) holds that high-level concepts are encoded as directions in transformer representation spaces. Park et al. [2024] formalize three notions of linearity—subspace, measurement, and intervention—and unify them via a causal inner product, demonstrating linear structure for 26 of 27 tested concepts in GPT-J. Marks and Tegmark [2024] show that truth and falsehood are linearly represented in LLaMA-13B, and that mass-mean probes (the difference-in-means direction) are more causally meaningful than logistic regression probes, with normalized indirect effects of 0.85–0.97 versus 0.33–0.52. We adopt the mass-mean probing methodology of Marks and Tegmark [2024] and apply it to a novel concept: Pokémon franchise membership. Unlike truth or language identity, this category is culturally constructed and has inherently fuzzy boundaries, making it a more challenging test of the LRH.

Representation engineering. Zou et al. [2023] introduce Representation Engineering (RepE), a top-down approach that uses contrastive stimuli to extract concept directions and then reads or steers model behavior along these directions. Turner et al. [2023] propose Activation Addition (ActAdd), which steers model behavior by adding contrastive activation vectors to the residual stream during inference. Both approaches demonstrate that linear directions can causally influence model outputs. Our work focuses on the probing (reading) side—we extract and validate a Pokémon direction but leave causal steering experiments to future work.

Factual knowledge in transformers. Meng et al. [2022] show that factual associations in GPT-2 are stored in middle-layer MLP modules and can be edited with rank-one updates. Their causal tracing technique reveals that the last subject token position is critical for factual recall. We follow this finding by extracting activations at the last token of each entity name. If “is a Pokémon” functions partly as a factual association, we would expect it to be concentrated in similar locations—and indeed, we find the strongest signal in middle layers.

Sparse and nonlinear features. Cunningham et al. [2023] show that sparse autoencoders can decompose model activations into interpretable features, some corresponding to specific entity types. Engels et al. [2024] demonstrate that not all features are one-dimensional: some concepts (e.g., days of the week) are represented on multi-dimensional manifolds. These findings suggest that Pokémon-ness could, in principle, be a multi-dimensional feature encoding subtypes (fire, water, etc.) or generations. Our results show that even a single linear direction captures substantial structure, but we acknowledge that richer representations may exist in higher-dimensional subspaces.

3 Methodology

3.1 Task Formulation

We frame the task as a binary concept detection problem. Given an entity e , we ask whether the model’s internal representation encodes the concept “ e is a Pokémon.” Following the LRH, we hypothesize that this concept corresponds to a direction $\theta \in \mathbb{R}^d$ in the residual stream, such that projecting an entity’s activation onto θ yields a scalar score reflecting the entity’s degree of Pokémon-ness.

3.2 Data

Pokémon entities. We collect 1,026 Pokémon names spanning Generations 1–9 from the `pokemonData` repository.¹ We randomly sample 160 for training the probe direction and reserve 40 for testing.

Non-Pokémon entities. We curate 177 non-Pokémon entities across seven categories designed to span a range of expected Pokémon similarity: real animals (53), mythical creatures (30), Digimon (22), Pokémon-like invented names (20), other game creatures (17), fictional characters (15), and common objects (20). We split these 70/30 into 122 training and 55 test entities, stratified by category.

Prompt template. Each entity is embedded in the template “{entity} is a” and tokenized with a beginning-of-sequence token prepended. We extract activations at the last token position of the entity name, following the finding of Meng et al. [2022] that the last subject token is critical for factual recall.

3.3 Mass-Mean Probe

We compute the Pokémon direction at each layer ℓ as the normalized difference in means between Pokémon and non-Pokémon training activations:

$$\theta_\ell = \frac{\mu_\ell^+ - \mu_\ell^-}{\|\mu_\ell^+ - \mu_\ell^-\|}, \quad (1)$$

where $\mu_\ell^+ = \frac{1}{|\mathcal{D}^+|} \sum_{e \in \mathcal{D}^+} \mathbf{a}_\ell(e)$ is the mean activation over Pokémon training entities and μ_ℓ^- is defined analogously over non-Pokémon training entities. The Pokémon quality score for any entity e is then:

$$s_\ell(e) = \theta_\ell^\top \mathbf{a}_\ell(e). \quad (2)$$

We classify an entity as Pokémon if $s_\ell(e)$ exceeds a threshold τ , set as the midpoint between the training class means projected onto θ_ℓ .

3.4 Model and Implementation

We use GPT-2-XL (1.5B parameters, 48 layers, $d = 1600$) loaded via TRANSFORMERLENS [Nanda and Bloom, 2022]. We extract residual stream activations at every second layer (24 layers total) in float16 precision with batch size 16. The entire pipeline—activation extraction plus analysis—runs in approximately 15 seconds on a single NVIDIA RTX A6000 GPU. The mass-mean probe is deterministic and has no stochastic components, so a single run suffices.

3.5 Baselines

We compare the mass-mean probe against two baselines:

- **Logistic regression:** A supervised linear classifier trained on the same train split activations at the best layer. This has strictly more capacity than the mass-mean direction (a full weight vector rather than the difference in means).
- **Random direction:** A randomly sampled unit vector in $\mathbb{R}^d$, providing a chance-level null baseline.

¹<https://github.com/lgreski/pokemonData>

Table 1: Layer-wise classification performance of the mass-mean probe. Best accuracy and AUC in **bold**. The Pokémon direction peaks in middle layers (10–32) and degrades in the final layers.

Layer	Test Accuracy	AUC	Cohen’s d
0	0.768	0.828	1.54
4	0.811	0.885	1.94
8	0.853	0.919	1.91
10	0.884	0.934	1.82
12	0.884	0.938	1.84
16	0.853	0.951	2.15
26	0.863	0.955	2.52
30	0.884	0.960	2.59
32	0.884	0.965	2.62
38	0.811	0.930	2.58
44	0.821	0.866	2.35
46	0.811	0.806	1.78

Table 2: Method comparison at layer 10. The unsupervised mass-mean probe outperforms supervised logistic regression. Best results in **bold**.

Method	Test Accuracy	AUC
MASS-MEAN (ours)	0.884	0.934
LOGISTIC REGRESSION	0.832	0.941
RANDOM DIRECTION	0.505	—

3.6 Evaluation

We evaluate using classification accuracy and area under the ROC curve (AUC) on the held-out test set (40 Pokémon, 55 non-Pokémon). We report Cohen’s d as an effect size measure for the separation between Pokémon and non-Pokémon projections. For the non-Pokémon ranking analysis, we use the Kruskal-Wallis H test to assess whether Pokémon quality scores differ significantly across entity categories.

4 Results

4.1 The Pokémon Direction Is Linearly Encoded

The mass-mean probe achieves 88.4% test accuracy at layer 10 (AUC = 0.934), confirming that “is a Pokémon” is linearly encoded in GPT-2-XL’s residual stream. Table 1 shows classification performance across layers. The Pokémon direction emerges as early as layer 4 (81.1% accuracy) and maintains strong performance through layers 10–32, with the highest AUC of 0.965 at layer 32. Performance degrades in the final layers (44–46), consistent with prior findings that conceptual representations are strongest in middle transformer layers [Marks and Tegmark, 2024, Meng et al., 2022].

4.2 Mass-Mean Outperforms Logistic Regression

Table 2 compares the mass-mean probe against baselines at layer 10. The mass-mean probe outperforms logistic regression in accuracy (88.4% vs. 83.2%) despite being unsupervised. The random direction baseline performs at chance (50.5%), confirming that the Pokémon direction captures genuine structure rather than an artifact of high-dimensional geometry.

4.3 Non-Pokémon Entities Form a Spectrum of Pokémon Quality

Projecting all 177 non-Pokémon entities onto the Pokémon direction at layer 10 reveals a continuous distribution of Pokémon quality scores, not a binary split. Table 3 shows the 15 highest-scoring non-Pokémon entities. Thirteen non-Pokémon entities cross the classification boundary: four invented

Table 3: Top 15 non-Pokémon entities by Pokémon quality score at layer 10. Entities above the dashed line were classified as Pokémon by the probe. Game creatures from Nintendo franchises and invented monster names dominate the rankings.

Rank	Entity	Category	Score	Classified as Pokémon?
1	Crystalix	Pokémon-like names	−10.13	Yes
2	Cactuar	Other game creatures	−10.76	Yes
3	Frostbeak	Pokémon-like names	−11.40	Yes
4	Koopa	Other game creatures	−11.60	Yes
5	Thunderix	Pokémon-like names	−12.72	Yes
6	Pianta	Other game creatures	−13.50	Yes
7	Samus	Fictional characters	−13.71	Yes
8	Goomba	Other game creatures	−14.34	Yes
9	Garurumon	Digimon	−14.87	Yes
10	Mistral	Pokémon-like names	−15.06	Yes
11	armadillo	Real animals	−15.56	Yes
12	Kirby	Other game creatures	−15.83	Yes
13	Moogles	Other game creatures	−15.86	Yes
14	Boulderback	Pokémon-like names	−16.58	No
15	kitsune	Mythical creatures	−16.80	No

Table 4: Mean Pokémon quality score by entity category at layer 10, sorted by score. Higher (less negative) scores indicate greater Pokémon-likeness. Differences are significant at $p < 10^{-10}$ (Kruskal-Wallis).

Category	Mean Score	Std Dev	N
Pokémon-like names	−21.58	8.59	20
Other game creatures	−22.12	11.67	17
Digimon	−27.20	5.67	22
Mythical creatures	−32.90	14.58	30
Real animals	−36.81	16.42	53
Fictional characters	−38.59	15.07	15
Common objects	−54.71	12.23	20

Pokémon-sounding names (Crystalix, Frostbeak, Thunderix, Mistral), six game creatures (Cactuar, Koopa, Pianta, Goomba, Kirby, Moogles), one Digimon (Garurumon), one real animal (armadillo), and one fictional character (Samus).

The bottom of the ranking is equally informative. The least Pokémon-like entities are common real animals (horse at −79.98, frog at −75.45, dog at −74.72) and common objects. Notably, “dragon” scores −74.64 despite many dragon-type Pokémon existing, suggesting the model distinguishes between the general concept of a dragon and dragon-type Pokémon.

4.4 Category-Level Analysis

Table 4 shows mean Pokémon quality scores by category. The ordering is highly interpretable: invented Pokémon-sounding names and game creatures from similar franchises are most Pokémon-like, while common objects are maximally distant. Digimon rank third—close to Pokémon but clearly distinguished, suggesting the model encodes franchise-specific knowledge rather than generic “fictional creature-ness.” The Kruskal-Wallis test confirms that these differences are highly significant ($H = 58.22, p = 1.03 \times 10^{-10}$).

4.5 Pokémon Quality Varies Among Actual Pokémon

The Pokémon direction also distinguishes among actual Pokémon. Iconic or distinctively named Pokémon like Gothorita (−5.2) and Oshawott (−6.8) score higher than obscure ones like Silcoon (−31.5) and Clobbopus (−28.6). This suggests the direction captures not just category membership but also how “recognizably Pokémon” a name sounds—a reflection of the entity’s prominence in the model’s training data.

5 Discussion

Extending the Linear Representation Hypothesis. Our results demonstrate that the LRH extends beyond clean binary concepts like truth and sentiment to culturally constructed categories with fuzzy boundaries. The concept “is a Pokémon” is not a natural kind—it is defined by a commercial franchise with over 1,000 members spanning nine generations of games. Yet a simple difference-in-means direction captures this category with 88.4% accuracy and large effect sizes (Cohen’s d up to 2.62). This suggests that linear representations are a general-purpose encoding mechanism in transformers, applicable even to arbitrary cultural categories learned from training data.

Continuous rather than binary representations. A striking finding is that the model does not encode a sharp binary boundary between Pokémon and non-Pokémon. Instead, entities fall along a continuous spectrum, with the ranking matching human intuitions: game creatures from similar franchises are “almost Pokémon,” while chairs and computers are “maximally not Pokémon.” This graded representation is consistent with prototype theory in cognitive science [Rosch, 1975], where category membership is a matter of degree rather than a sharp boundary.

What drives Pokémon quality? Three factors likely contribute to an entity’s Pokémon quality score. First, *morphological similarity*: names with Pokémon-like phonotactics (e.g., Crystalix, Thunderix) score high regardless of whether they refer to real entities. Second, *semantic similarity*: creatures from similar game franchises co-occur with Pokémon in training data, leading to similar representations. Third, *franchise co-occurrence*: Samus, a Nintendo character who is not creature-like at all, scores high—likely because Nintendo franchise content frequently appears alongside Pokémon content in the training corpus. Disentangling these factors is an important direction for future work.

Limitations. Our study has several limitations. (1) We analyze only GPT-2-XL; results may not generalize to other architectures or model scales. Larger models likely encode cleaner representations. (2) We use a single prompt template (“{entity} is a”); other templates may activate different aspects of entity knowledge. (3) We do not perform causal validation—we have not verified that adding the Pokémon direction to non-Pokémon activations produces Pokémon-like model outputs. Without this, the direction may be correlated with but not causally responsible for Pokémon-related behavior. (4) The non-Pokémon test set is small (55 examples), yielding wide confidence intervals. (5) High Pokémon quality scores for some entities may reflect training data co-occurrence rather than genuine conceptual similarity—the model may have seen these entities discussed alongside Pokémon on gaming forums or wikis.

Broader implications. Our findings have implications for creative AI applications. A model that encodes “Pokémon quality” as a continuous, manipulable direction could, in principle, be steered to generate novel Pokémon-like entities by adding the Pokémon direction to the residual stream during generation [Turner et al., 2023]. More broadly, the ability to extract and rank entities along culturally defined axes opens possibilities for analyzing how models organize knowledge about brands, franchises, genres, and other human-constructed categories.

6 Conclusion

We show that the concept “X is a Pokémon” is encoded as a linear direction in GPT-2-XL’s residual stream, concentrated in middle layers (10–32). A mass-mean probe achieves 88.4% classification accuracy on held-out entities, outperforming both logistic regression and random baselines. By projecting non-Pokémon entities onto this direction, we discover a continuous, interpretable spectrum of Pokémon quality: game creatures from similar franchises and invented monster names score highest, Digimon rank in the middle, and common objects are maximally distant.

The key takeaway is that linear representations in language models extend to fine-grained, culturally constructed categories—not just broad semantic properties—and that these representations encode graded similarity rather than binary membership.

Future work should validate these findings with causal intervention experiments, replicate across larger models, and investigate whether sub-categories (Pokémon types, generations) form additional structure within the Pokémon subspace. Sparse autoencoder decomposition may reveal whether a discrete “Pokémon feature” exists alongside the linear direction we identify.

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